

Maru Metall — Final Competition Package (v2)

28DIGITAL Business Challenge 2026 * EIT Digital

Challenge: AI-Supported Cost Estimation Tool

IMPORTANT — v2 replaces the previous version entirely. The previous version incorrectly addressed the generic “Challenges - 28DIGITAL.pdf” brief (circular economy, GTM, BMC for a new service). This version correctly addresses the **specific Maru Metall brief** found in “28DIGITAL Business Challenges - Maru Metall presentation.pdf”: **Build an AI-supported cost estimation tool** that turns diverse project documentation into a reliable quotation.

What the actual challenge asks for

From the official Maru Metall brief (6-slide PDF):

Task	Description
01 Input Interpretation	Identify fabrication scope from mixed-quality drawings, models, lists and specifications
02 Quantity & Effort	Extract material quantities; estimate fabrication labour, surface treatment, subcontracting, packaging, transport
03 Risk & Assumptions	Make missing information, assumptions, exclusions and technical/commercial risks explicit
04 Output	Consistent, traceable cost estimate that an experienced estimator can review and adjust

Success dimensions: Traceability * Human control * Variable input handling * Scalability

Expected deliverable: A **proof of concept** to support a go/no-go decision — NOT a fully functional tool.

File Index

File	Purpose
01_pitch_script.md	3-minute pitch script, 4 speakers, exact dialogue + timing + delivery notes
02_business_model_canvas.md	Full 9-block BMC for SteelQuote AI (internal tool + optional SaaS)
03_unit_economics.md	Financial model: hour recovery, project data (Skybridge + Akkasater), ROI table

File	Purpose
04_poc_design.md	System architecture, PoC walkthrough on Example 4, input quality handling, human controls
05_qa_defence.md	10 hardest jury questions: short answer + full answer, delivery tips
06_pitch_deck.html	Interactive 7-slide visual pitch deck (open in browser, navigate with ← ->)

Core Concept

SteelQuote AI — an AI-supported cost estimation tool that interprets any combination of IFC models, Excel BOMs, PDF drawings, and email specifications, applies Maru’s norm-based production logic, and produces a structured, traceable cost estimate that an experienced estimator can review, adjust, and send.

The four pipeline stages directly map to the four challenge tasks: 1. **Input Interpretation** -> document parser + LLM extraction 2. **Quantity & Effort** -> norm-based labour engine + surface treatment pricing 3. **Risk & Assumptions** -> explicit flagging layer, assumption log 4. **Output** -> structured line-item estimate in Maru’s standard format

Key Numbers

Metric	Value
Annual inquiries at Maru	500+
Engineering hours/year (est.)	~6,000 h
Annual estimation cost (est.)	~€330,000
Hours recovered (base, 60%)	1,800 h/year
Annual value recovered	€99,000/year
PoC build cost	€13,000–€25,000
Payback period	2–3 months
Reference project accuracy (PoC)	–0.4% vs. actual quote

Real Project Data Used

Project	Weight	Total Value	Reference
Pedestrian Steel Skybridge (HP11257-03)	206,414 kg	€591,285	Example 4 workbook + actual Maru quote

Project	Weight	Total Value	Reference
Akkasater Storage Hall (TU4031)	148,315 kg	€333,395	Example 2 workbook + actual Maru quote

Evaluation Criteria Mapping

Criterion (25% each)	Where covered
Problem & Value Proposition	Pitch slide 1, Unit economics, BMC VP
Business Model & Viability	BMC all 9 blocks, unit economics ROI
Go-to-Market & Feasibility	PoC design (internal deploy first), pitch slide 4
Pitch Quality & Q&A	Script with delivery notes, Q&A playbook